Descriptive Analytics Report: Zepto

Product Catalog & Pricing Behavior at Scale

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Course: BCA (DS & AI)

Section: BCADS24

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Tracking Catalog Composition, Pricing, Discounts, Stock Availability & Assortment Structure

Note on data: This report uses the supplied Zepto product-catalog dataset containing 3,732 product records across 14 categories. The dataset is inventory/catalog-level data (product, price, discount, stock status) rather than raw event-level order or sales transaction data. 6 pairs of category labels are duplicated across identical underlying rows (Cooking Essentials/Munchies, Packaged Food/Ice Cream & Desserts/Chocolates & Candies, Dairy Bread & Batter/Beverages, Personal Care/Paan Corner), so category-level charts in this report show matching bars for those label pairs. Therefore, the analysis describes the supplied catalog observations and does not assume a larger real-time sales or order stream.

1. Objective

Build a descriptive analytics layer over the Zepto product-catalog data to answer the following recurring business questions:

- How are products distributed across categories?
- Which categories carry the highest average selling price?
- Which categories offer the highest average discounts?
- What is the overall stock availability status of the catalog?
- Which categories show the highest out-of-stock rates?
- How are selling prices distributed across the catalog?
- What is the relationship between MRP and discounted selling price?
- Which products offer the deepest discounts?
- How is product weight distributed across the catalog?
- How does the catalog split across price bands and discount bands?
- Which products carry the deepest discounts, and which are the most expensive?
- How does product weight relate to price and category?
- What is the on-hand inventory value, and how many products are at risk of stocking out?
- What does an overall catalog-health summary look like?

2. How This Would Work at Real Scale

Processing a large, constantly-changing product catalog is an engineering problem before it is an analytics problem. A realistic architecture would ingest catalog/price-feed updates, aggregate them by category and SKU, store raw and processed data, and expose the resulting KPIs through a data-visualization layer.

Layer	Purpose	Typical tools
Investigation	Research, understand and prepare the catalog dataset	Google, Claude & ChatGPT
Stream processing	Aggregate by category, price band and stock status	Jupyter Notebook (Python)
Storage	Retain raw and processed catalog data	Excel files / CSV files
Batch aggregation	Daily/periodic KPI and summary metrics	Jupyter Notebook
Report	Present analysis, graphs and insights	Data Visualization / MS Word

3. Category-wise Product Distribution

Across the supplied catalog, the dataset records 3,732 products spanning 14 categories. The chart below shows how the product assortment (SKU count) is distributed across categories.

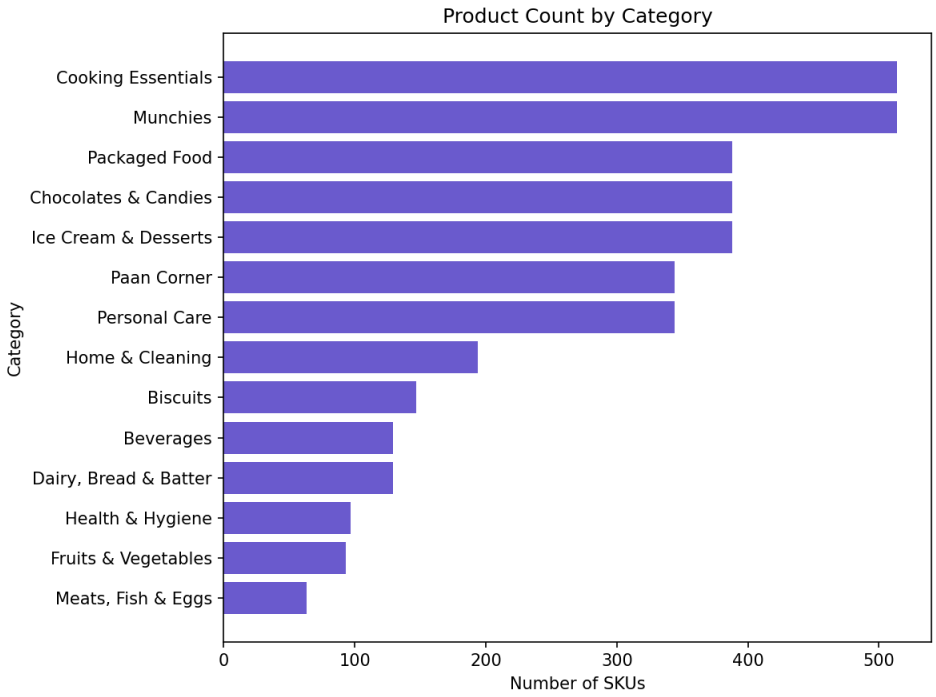


Figure 1. Product count by category

INSIGHTS:

- Munchies and Cooking Essentials are the largest categories by SKU count, each with 514 products.
- Meats, Fish & Eggs and Fruits & Vegetables have the smallest assortments, with 63 and 93 products respectively.
- A wide gap between the top and bottom categories suggests the catalog is weighted toward packaged / long-shelf-life goods over fresh produce.
- Category-level SKU counts can help guide catalog-expansion and de-listing decisions.

KPI:

Total products: 3,732
Total categories: 14
Largest category: Cooking Essentials / Munchies (514 SKUs each)
Smallest category: Meats, Fish & Eggs (63 SKUs)

4. Average Selling Price by Category

The overall average selling price across the supplied catalog is ~₹142. The chart below shows how average selling price varies by category.



Figure 2. Average selling price by category

INSIGHTS:

- Paan Corner and Personal Care carry the highest average selling price (~₹190) in the supplied data.
- Fruits & Vegetables and Biscuits carry the lowest average selling price, consistent with smaller pack sizes and everyday-use pricing.
- Price positioning varies meaningfully by category, which can inform category-level pricing and promotion strategy.

KPI:

Overall avg. selling price: ≈ ₹142
Overall avg. MRP: ≈ ₹157
Highest-priced category: Paan Corner / Personal Care (≈ ₹190)
Lowest-priced category: Fruits & Vegetables (≈ ₹40)

5. Average Discount % by Category

The overall average discount across the supplied catalog is ≈7.6%. The chart below compares average discount percentage across categories.

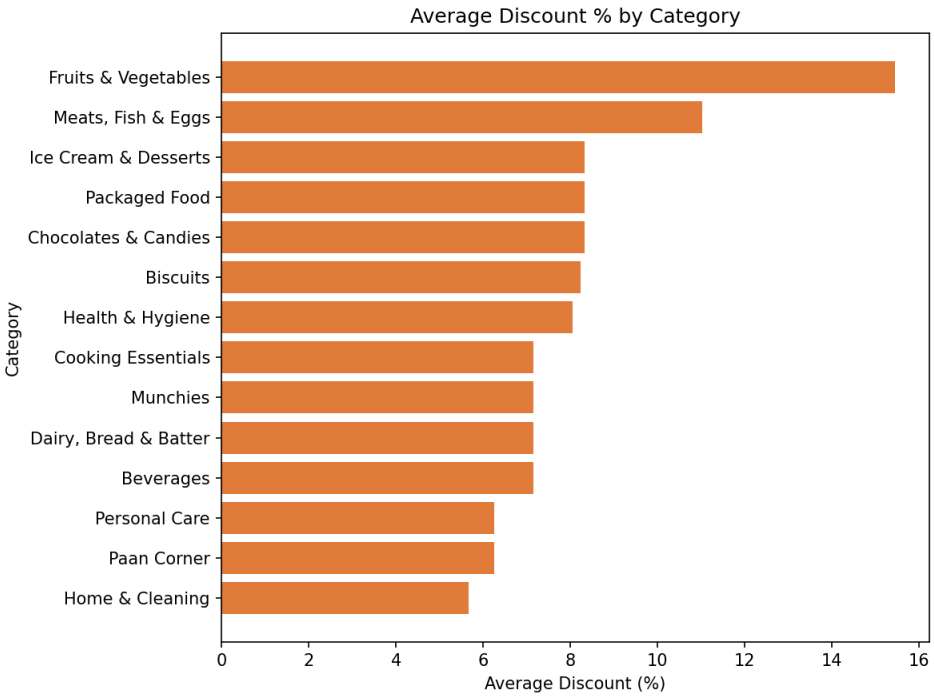


Figure 3. Average discount percentage by category

INSIGHTS:

- Fruits & Vegetables shows the highest average discount (≈15.5%), consistent with the category's perishability and need for faster turnover.
- Meats, Fish & Eggs is the second-most discounted category, also a perishable/fast-turnover category.
- Home & Cleaning shows the lowest average discount (≈5.7%), suggesting more stable, less promotion-driven pricing.
- Discount depth appears linked to product perishability rather than being uniform across the catalog.

KPI:

Overall avg. discount: ≈ 7.6%
Highest-discount category: Fruits & Vegetables (≈ 15.5%)
Lowest-discount category: Home & Cleaning (≈ 5.7%)
Total estimated savings vs. MRP: ≈ ₹55,505 across the catalog

6. Out-of-Stock Rate by Category

Of the 3,732 supplied products, 453 (≈12.1%) are recorded as out of stock. The chart below breaks this down by category.

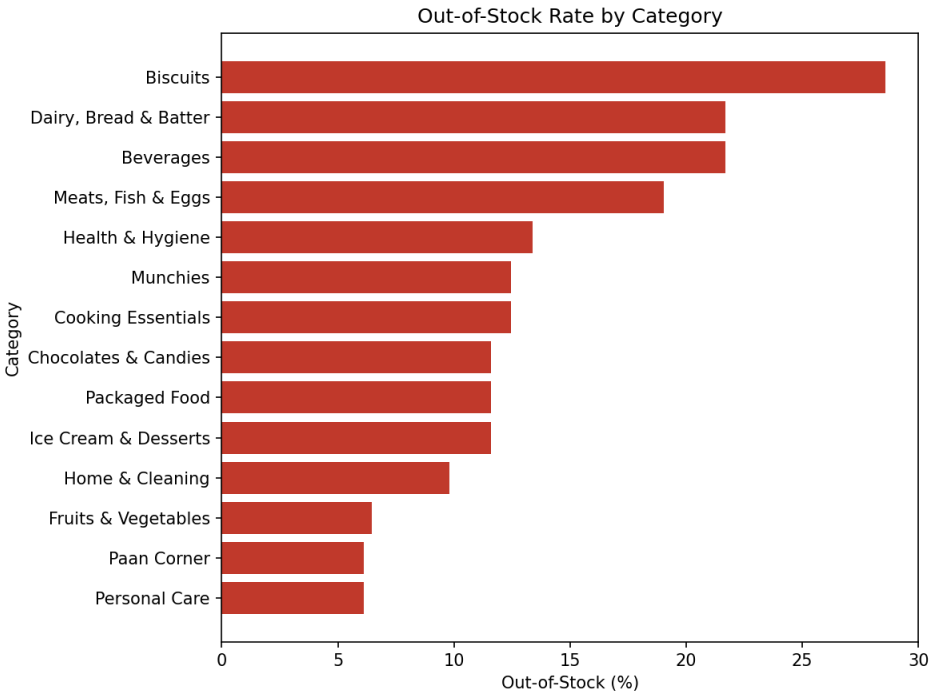


Figure 4. Out-of-stock rate by category

INSIGHTS:

- Biscuits shows the highest out-of-stock rate in the supplied data, at ≈28.6%.
- Beverages and Dairy, Bread & Batter follow with elevated out-of-stock rates.
- Personal Care and Paan Corner show the lowest out-of-stock rates (≈6.1%), indicating comparatively stronger stock availability.
- Categories with high out-of-stock rates may need closer supply-chain or replenishment attention.

KPI:

Overall out-of-stock rate: ≈ 12.1% (453 of 3,732 products)
Highest out-of-stock category: Biscuits (≈ 28.6%)
Lowest out-of-stock category: Personal Care / Paan Corner (≈ 6.1%)

7. Overall Stock Availability

Viewed at the catalog level (rather than by category), the majority of products in the supplied dataset are currently in stock.

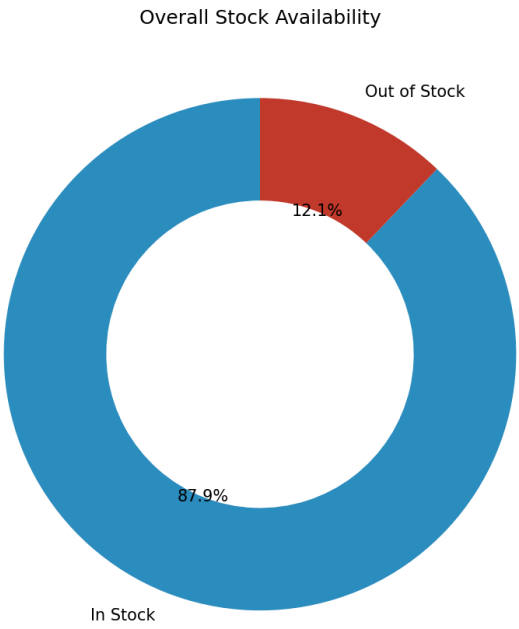


Figure 5. Overall stock availability

INSIGHTS:

- ≈87.9% of the supplied catalog is in stock, while ≈12.1% is out of stock.
- A double-digit out-of-stock share, even if not extreme, represents lost potential sales and should be monitored.
- This KPI is best tracked over time rather than as a single-point snapshot.

KPI:

In stock: 3,279 products (≈ 87.9%)
Out of stock: 453 products (≈ 12.1%)

8. Selling Price Distribution

The histogram below shows how selling prices are distributed across the full catalog.

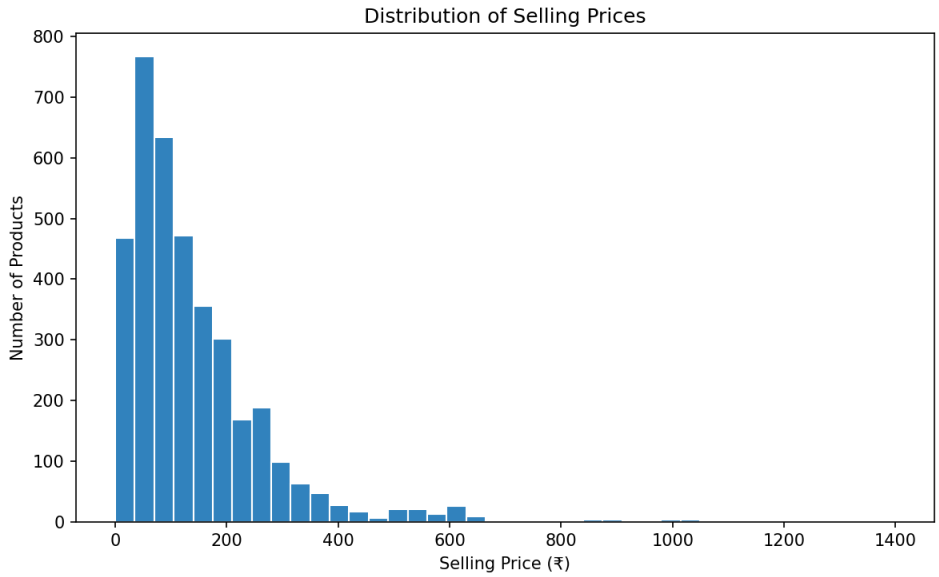


Figure 6. Distribution of selling prices

INSIGHTS:

- The distribution is right-skewed: most products are priced under ₹200, with a long tail of higher-priced items.
- The median selling price (≈₹104) is noticeably lower than the mean (≈₹142), confirming the right-skew.
- This pattern is typical of a grocery/quick-commerce catalog, where everyday essentials dominate by count.

KPI:

Median selling price: ≈ ₹104
Mean selling price: ≈ ₹142
Maximum selling price: ≈ ₹1,399 (Borges Extra Light Olive Oil Bottle)

9. Product Distribution by Price Band

To make the price distribution easier to act on, products are grouped into four price bands: Budget (under ₹50), Value (₹50-150), Mid (₹150-300), and Premium (above ₹300).

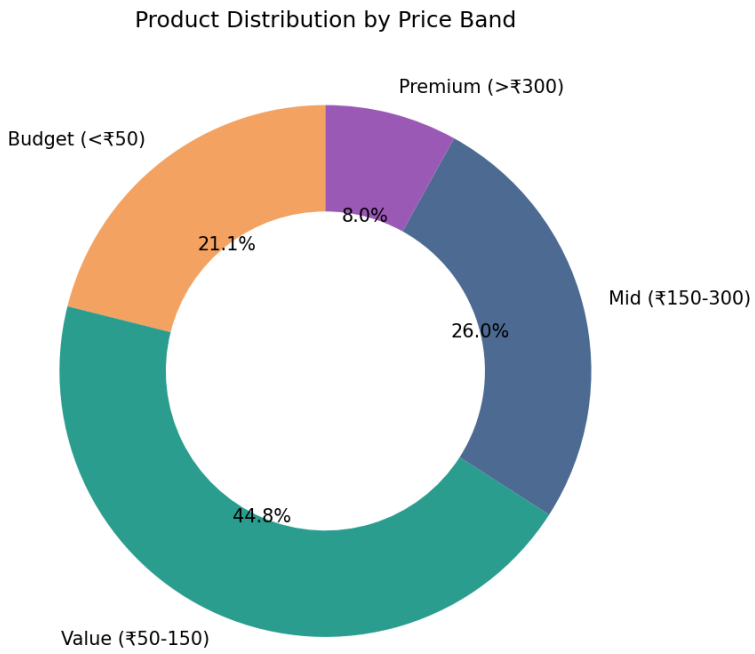


Figure 7. Product distribution by price band

INSIGHTS:

- The Value (₹50-150) band is the largest, holding 1,673 products — ≈44.8% of the catalog.
- Budget items (under ₹50) make up a significant share at ≈21.1% (787 products), confirming the catalog's everyday-essentials focus.
- Premium items (above ₹300) are a minority at ≈8.0% (300 products), suggesting limited high-ticket assortment.
- This banding can support tiered promotional strategy — e.g., bundling Value-band items to lift average order value.

KPI:

Largest price band: Value (₹50-150) — 1,673 products, 44.8%
Smallest price band: Premium (>₹300) — 300 products, 8.0%

10. Product Distribution by Discount Band

Products are further grouped by discount depth: No discount (0%), Low (1-10%), Medium (11-25%), and High (above 25%).

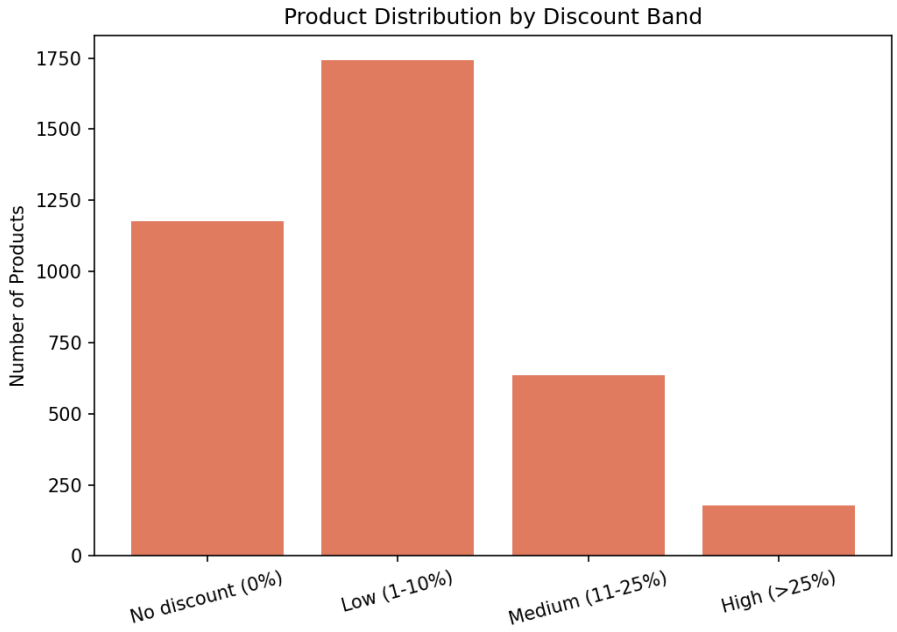


Figure 8. Product distribution by discount band

INSIGHTS:

- Low discounts (1-10%) are the most common band, covering 1,742 products (≈46.7% of the catalog).
- A substantial 1,178 products (≈31.6%) carry no discount at all — full MRP pricing.
- Only 178 products (≈4.8%) fall into the High-discount band (above 25%), indicating deep discounting is the exception rather than the rule.
- This distribution suggests headroom for more promotional activity if the business wants to compete more aggressively on price.

KPI:

No-discount products: 1,178 (≈ 31.6%)
High-discount products (>25%): 178 (≈ 4.8%)

11. MRP vs Selling Price Relationship

The scatter plot below relates each product's MRP to its discounted selling price, colored by discount percentage, to show how discounting behaves across the price range.

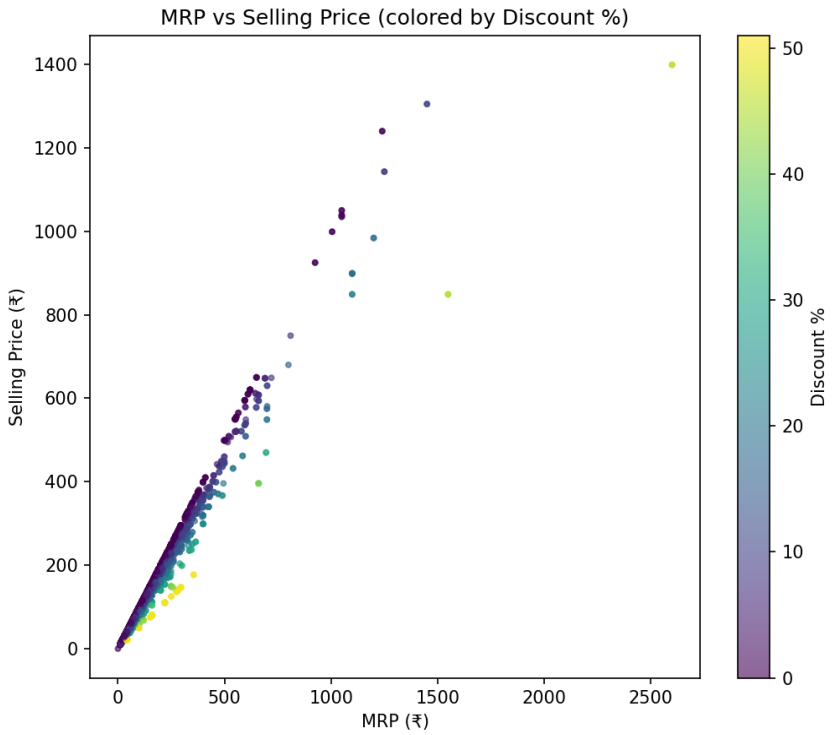


Figure 9. MRP vs selling price, colored by discount %

INSIGHTS:

- Selling price rises broadly in line with MRP, as expected, but the colour gradient shows discount depth is not uniform at a given price point.
- Points that fall further below the diagonal represent products with deeper discounts relative to their MRP.
- This view helps identify where promotional intensity is concentrated across the price spectrum.

12. Top 10 Highest-Discount Products

The table below lists the ten products with the deepest discounts in the supplied catalog.

Product	Category	MRP (₹)	Selling Price (₹)	Discount
Dukes Waffy Orange Wafers	Biscuits	45	22	51%
Dukes Waffy Strawberry Wafers	Biscuits	45	22	51%
Dukes Waffy Chocolate Wafers	Biscuits	45	22	51%
Dukes Waffy Strawberry Roll	Biscuits	150	75	50%
Moi Soi Manchurian Sauce	Packaged Food	280	140	50%
Chef's Basket Elbow Pasta	Cooking Essentials	160	80	50%
RRO Mozzarella Pizza Cheese	Dairy, Bread & Batter	275	137	50%
Epigamia Fruit Yogurt Vanilla	Health & Hygiene	40	20	50%
Epigamia Fruit Yogurt Strawberry	Health & Hygiene	40	20	50%
Chef's Basket Penne Pasta	Cooking Essentials	160	80	50%

INSIGHTS:

- Biscuits (Dukes Waffy range) tops the discount list at 51% off MRP, the highest in the supplied catalog.
- Several dairy/packaged-food and health items also feature deep 50% discounts, suggesting active clearance or promotional pricing across those segments.
- Deep-discount products are concentrated in a handful of categories rather than spread evenly across the catalog.

13. Top 10 Most Expensive Products

For context alongside the deepest discounts, the table below lists the ten highest-selling-price products recorded in the catalog.

Product	Category	MRP (₹)	Selling Price (₹)	Discount
Borges Extra Light Olive Oil Bottle	Cooking Essentials	2,600	1,399	46%
Praakritik Natural Desi Gir Cow A2 Ghee	Cooking Essentials	1,450	1,305	10%
Saffola Gold (Jar)	Cooking Essentials	1,240	1,240	0%
Dhara Kachi Ghani Mustard Oil Jar	Cooking Essentials	1,250	1,143	8%
Dhara Filtered Groundnut Oil (Jar)	Cooking Essentials	1,050	1,050	0%
Fortune Rice Bran Health Oil (Jar)	Cooking Essentials	1,050	1,039	1%
Dhara Filtered Groundnut Oil (Jar)	Cooking Essentials	1,050	1,035	1%
Fortune Soyabean Oil	Cooking Essentials	1,005	999	0%
Dhara Health Refined Sun Flower Oil Jar	Cooking Essentials	1,200	984	18%
Fortune Sunlite Refined Sunflower (Jar)	Cooking Essentials	925	925	0%

INSIGHTS:

- Cooking oils and ghee dominate the most-expensive list — Borges Olive Oil, Dhara and Fortune oil variants, and Praakritik A2 Ghee together account for the entire top ten.
- Several of the highest-priced items (Saffola Gold, Dhara Groundnut Oil, Fortune Sunflower/Soyabean Oil) carry 0% discount, unlike the deep-discount list in Section 12.
- Where discounts do appear on premium items, they are modest (0-18%) compared with the 50-51% seen on the deepest-discount everyday products — a split worth validating against real sales data.

14. Product Weight Distribution

The chart below shows how product weight (grams) is distributed across the catalog, capped at 2,000g for readability.

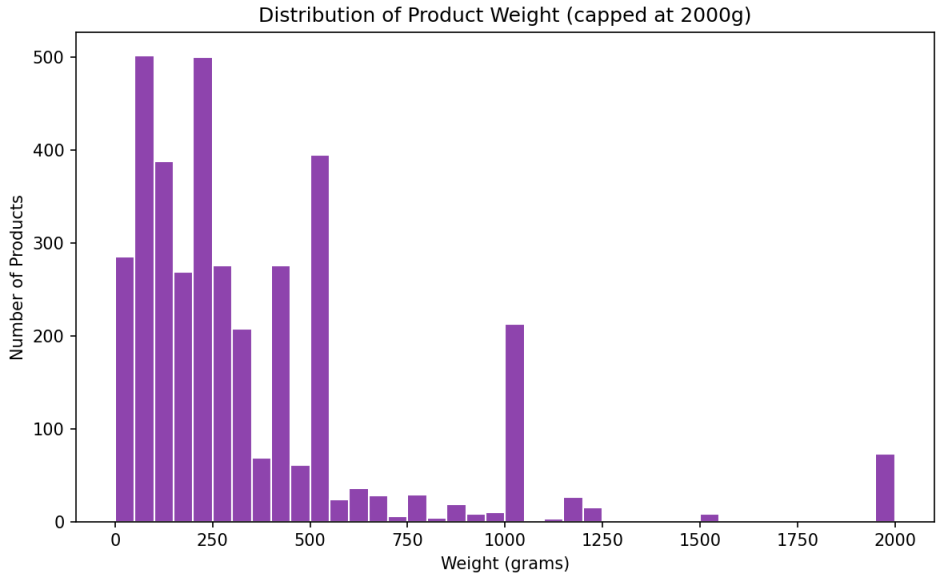


Figure 10. Distribution of product weight (capped at 2,000g)

INSIGHTS:

- Most products weigh under 500g, consistent with single-use or small household pack sizes typical of quick-commerce.
- A smaller number of bulkier items (above 1,000g) extend the tail of the distribution.
- Weight data can support logistics and packaging-cost planning alongside price and category data.

KPI:

Median product weight: 225 g
Mean product weight: ≈ 388 g
Maximum recorded weight: 10,000 g

15. Average Weight by Category

The chart below compares average product weight across categories, giving a sense of which categories skew toward bulkier packs.

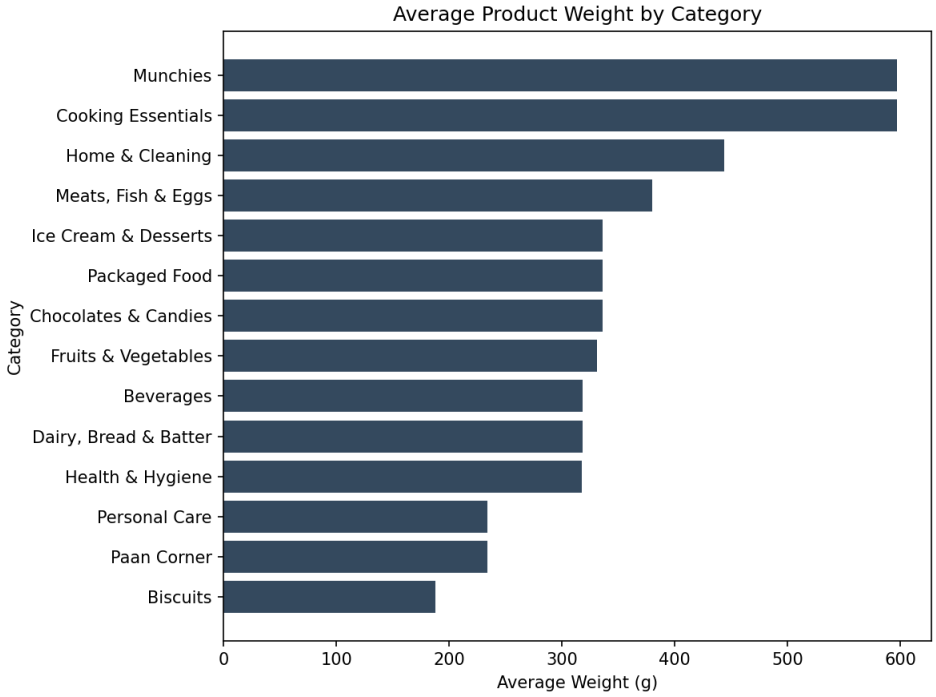


Figure 11. Average product weight by category

INSIGHTS:

- Cooking Essentials / Munchies carry the heaviest average pack size (≈597g), consistent with cooking oils, rice and staples sold in larger packs.
- Biscuits has the lightest average pack size (≈188g) in the supplied data, ahead of Paan Corner / Personal Care (≈234g).
- Weight, price and category are related: heavier categories also tend to carry higher average MRP (see correlation analysis in Section 16).

KPI:

Heaviest category (avg.): Cooking Essentials / Munchies (≈ 597 g)
Lightest category (avg.): Biscuits (≈ 188 g)

16. Correlation Between Numeric Fields

To understand how the catalog's numeric fields move together, the heatmap below shows pairwise correlation coefficients across MRP, discount %, available quantity, selling price, weight, and order-quantity unit.

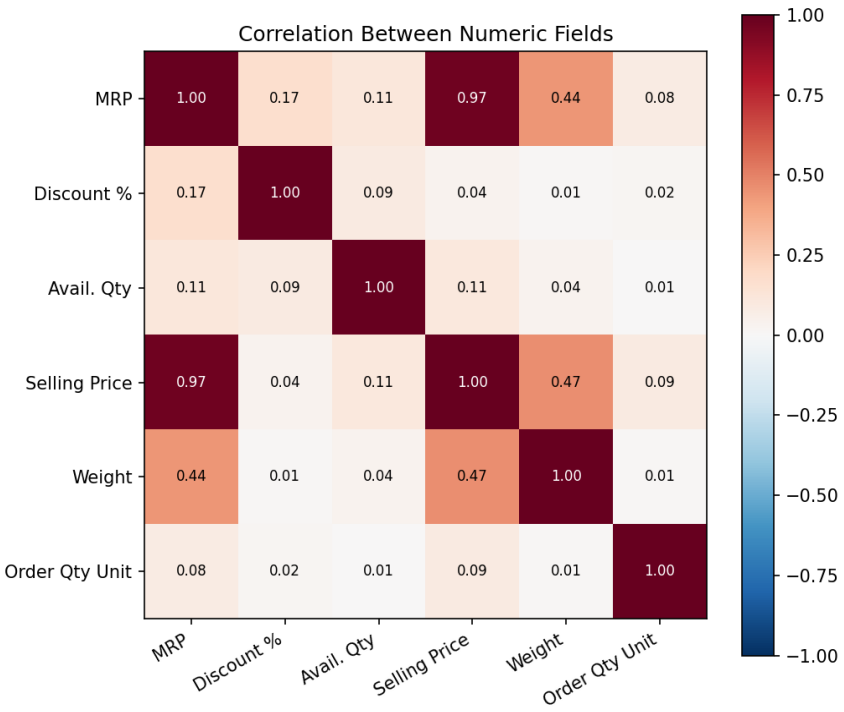


Figure 12. Correlation heatmap of numeric fields

INSIGHTS:

- MRP and Selling Price are almost perfectly correlated ($r \approx 0.97$), as expected since selling price is derived directly from MRP.
- Weight shows a moderate positive correlation with both MRP ($r \approx 0.44$) and Selling Price ($r \approx 0.47$) — heavier products tend to cost more, in line with the category patterns in Section 15.
- Discount % is only weakly correlated with MRP ($r \approx 0.17$) and essentially uncorrelated with weight ($r \approx 0.01$), suggesting discount decisions are not simply a function of product price or size.
- Available quantity shows weak correlation with all other fields, suggesting stock levels are set independently of price or discount strategy in this snapshot.

17. On-Hand Inventory Value by Category

Multiplying each product's available quantity by its MRP (and separately by its selling price) gives an estimate of on-hand inventory value — useful for understanding where capital is tied up in stock.

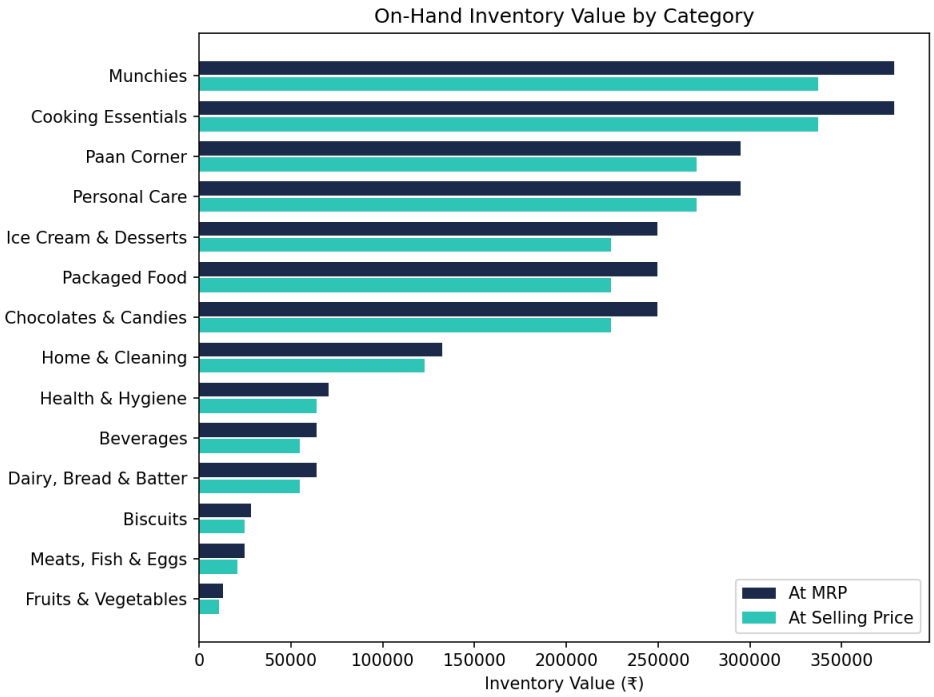


Figure 13. On-hand inventory value by category, at MRP vs. selling price

INSIGHTS:

- Cooking Essentials / Munchies holds the largest inventory value at both MRP and selling price, consistent with its large SKU count and higher average weight/price.
- The gap between MRP-value and selling-price-value across the catalog (≈₹2,49,700) represents the total markdown built into current listed prices.
- Fruits & Vegetables holds the smallest inventory value, reflecting both its small SKU count and low average price.

KPI:

Total inventory value at MRP: ≈ ₹24,92,781
Total inventory value at selling price: ≈ ₹22,43,081
Implied markdown exposure: ≈ ₹2,49,700

18. Low-Stock Products

Products with 1-3 units of available quantity (and not already flagged out of stock) represent near-term stock-out risk. The chart below shows how many such products exist per category.

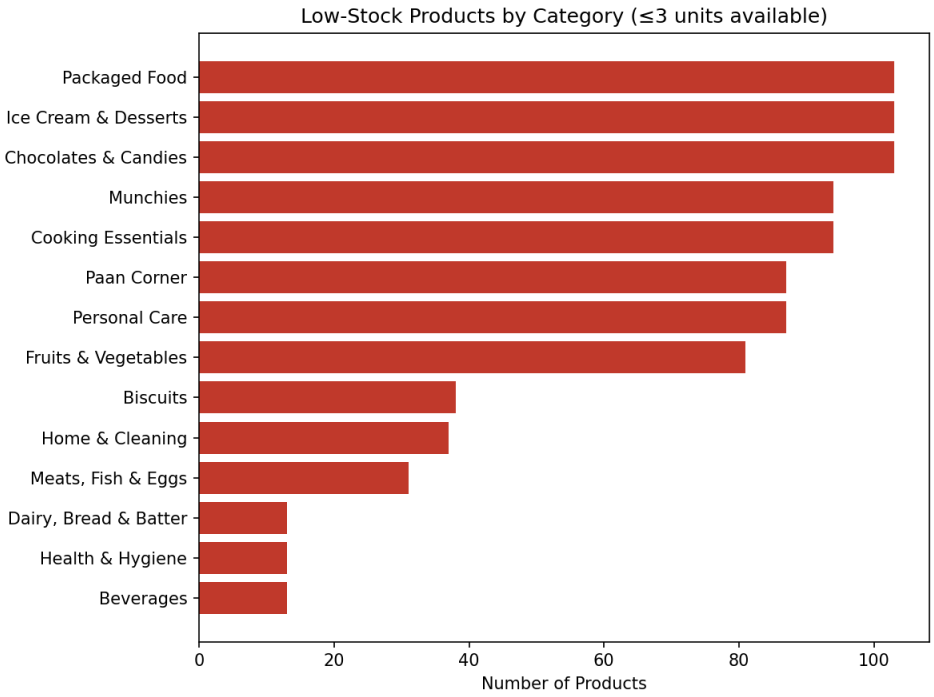


Figure 14. Low-stock products by category (≤ 3 units available)

INSIGHTS:

- 897 products — roughly 1 in 4 of the catalog — currently hold 3 or fewer units of available stock.
- Chocolates & Candies accounts for the largest share of low-stock products (103 SKUs), mirroring the large SKU counts of that category group.
- Beverages shows comparatively few low-stock items (13), suggesting steadier replenishment in that category.
- This list is a natural candidate for a proactive replenishment alert, since these SKUs are the most likely to tip into an out-of-stock state next.

KPI:

Products at risk of stock-out (≤3 units): 897 (≈ 24.0% of catalog)
Highest-risk category: Chocolates & Candies

19. Available Quantity Distribution

The histogram below shows how many units of stock are typically held per product, capped at 20 units for readability.

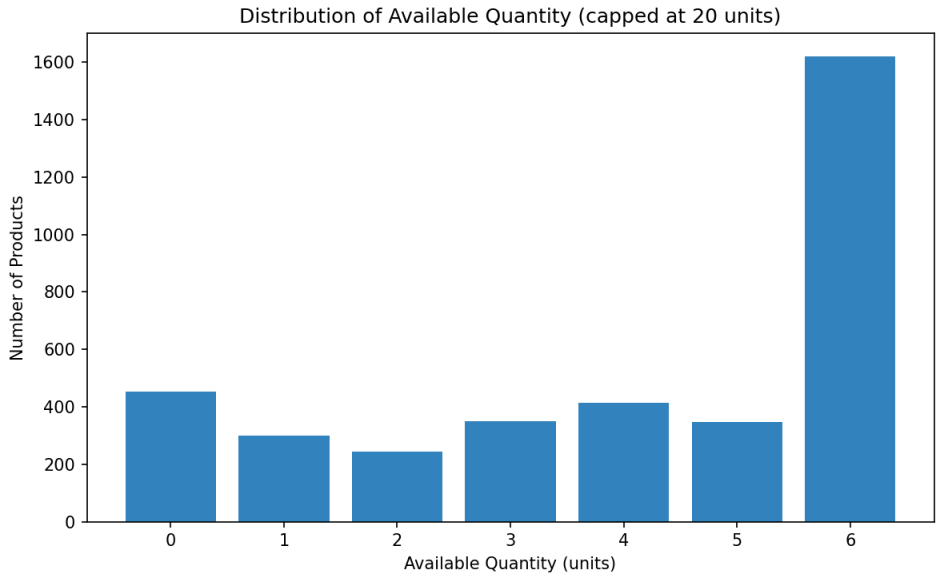


Figure 15. Distribution of available quantity per product

INSIGHTS:

- The median available quantity is 5 units per product, with a maximum of 6 units recorded in this snapshot — indicating tight, just-in-time stock holding rather than deep buffer inventory.
- A large cluster of products sit at 0-2 units, which overlaps heavily with the low-stock group flagged in Section 18.
- Consistently low per-SKU stock levels across the catalog suggest frequent replenishment cycles are needed to sustain availability.

KPI:

Median available quantity: 5 units
Mean available quantity: ≈ 4.0 units
Maximum available quantity: 6 units

20. Category Share of Total Catalog Value

Multiplying MRP by available quantity for every product and summing by category shows how total catalog value (at MRP) is distributed. This complements the per-unit pricing view in Section 4 with a stock-weighted view.

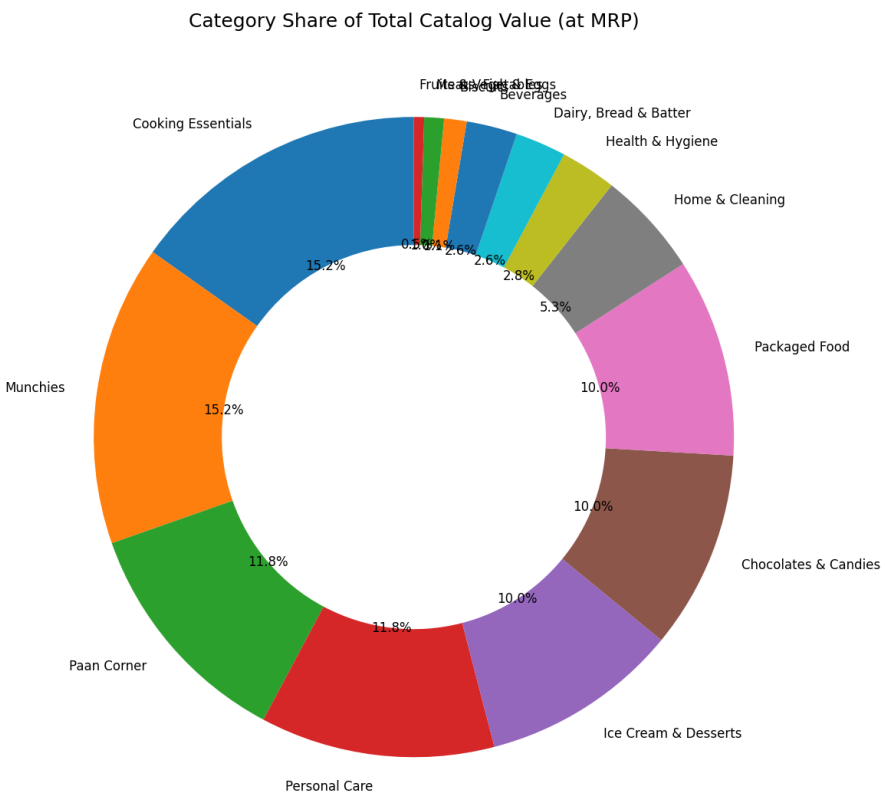


Figure 16. Category share of total catalog value, at MRP

INSIGHTS:

- Cooking Essentials and Munchies together account for the largest share of total catalog value, consistent with their high SKU count and above-average pack weight.
- Fruits & Vegetables and Meats, Fish & Eggs contribute the smallest shares of total catalog value, despite Meats, Fish & Eggs having a relatively high average price — its small SKU count limits its overall contribution.
- This stock-weighted view is useful for prioritising which categories matter most to overall inventory management, as opposed to per-product pricing alone.

Summary of Key Metrics

Metric	Value
Total products (SKUs)	3,732
Categories tracked	14
Overall avg. selling price	≈ ₹142
Overall avg. MRP	≈ ₹157
Overall avg. discount	≈ 7.6%
Total estimated savings vs. MRP	≈ ₹55,505
Out-of-stock products	453 (≈ 12.1%)
Low-stock products (≤3 units)	897 (≈ 24.0%)
Median product weight	225 g
Distinct underlying product sets	8 (behind 14 labels)
Total inventory value (MRP / Selling)	≈ ₹24.9L / ≈ ₹22.4L

Next Steps to Move from Descriptive → Predictive

- Demand/order forecasting by category once real sales/order-volume data is available.
- Stock-out prediction using historical availability patterns to flag at-risk SKUs before they go out of stock.
- Dynamic pricing analysis linking discount depth to sell-through, once transaction data is added.
- Assortment optimization using SKU-level performance once sales data is layered on top of catalog data.
- Trend and cohort analysis when longer historical catalog snapshots become available.

This report answers what the supplied catalog looks like today. The next analytical layer would combine this catalog data with order/transaction-level data to explain why stock or pricing patterns occur, and to forecast what is likely to happen next.

Glossary

MRP — Maximum Retail Price: the manufacturer-set ceiling price of a product before any discount is applied.

SKU — Stock Keeping Unit: a distinct, individually-tracked product/variant in the catalog.

Discount % — The percentage reduction from MRP to arrive at the discounted selling price shown to the customer.

Out of Stock — A product status flag indicating that available quantity is currently zero or unavailable for purchase.

Available Quantity — The recorded stock count for a product at the time the catalog snapshot was taken.

Category — A grouping of products by type (e.g., Biscuits, Beverages) used for catalog organization and reporting.